

미적분 평가과제

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$$10) -2r \cos 2\theta + 2 \int r \cos 2\theta \, dr = -2r \cos 2\theta + \sin 2\theta + C$$

$$(2) \quad -\frac{8}{5}(\cos 2\theta + \frac{1}{5}) \cos 2\theta \left[\frac{1}{2}(\cos \theta - \frac{8}{5} \cos 3\theta) \right]_0^{\frac{\pi}{2}} = \frac{16}{5} + \frac{1}{5}$$

$$(3) \quad \frac{1}{2} \ln 2 = \int_0^1 \frac{1}{2} x^0 dx = \left[\frac{1}{2} x^1 \ln x - \frac{1}{2} x^1 \right]_0^1 = \frac{3}{2} \ln 2 - \frac{1}{2} \cdot 2$$

$$(k): \int_2^4 x (\ln x)^2 dx = \frac{1}{2} x^2 (\ln x)^2 - \frac{1}{2} \int_2^4 x \ln x dx = \frac{1}{2} x^2 \ln x - \left[\frac{1}{2} x^2 \right]_2^4 = \frac{1}{2} x^2 \ln x - \frac{1}{2} x^2 \Big|_2^4$$

$$\frac{1}{2}x^2 + \frac{1}{2}x + \frac{1}{2} + c$$

$$(B): \text{Let } t = \cos \theta \Rightarrow -\sin \theta d\theta = dt \Rightarrow \int \frac{1}{t^2} dt = \int \frac{1}{t^2} dt \Rightarrow \frac{1}{t} + C = \frac{1}{\cos \theta} + C$$

$$= \frac{1}{\cos \theta} + C = \frac{1}{\cos \theta} + C = \frac{1}{\cos \theta} + C = \frac{1}{\cos \theta} + C$$

$$2. \int_{-1}^1 (m^2 - |x| + x^2) dx = 2 \int_0^1 (m^2 - |x| + x^2) dx = 2 \left[\frac{1}{2} m^2 x - \frac{1}{2} |x|^2 + \frac{1}{3} x^3 \right]_0^1 = 2 \left(\frac{1}{2} m^2 - \frac{1}{2} + \frac{1}{3} \right) = 2 \left(\frac{1}{2} m^2 - \frac{1}{6} \right) = m^2 - \frac{1}{3}$$

1. $\int_0^{\frac{\pi}{2}} 2 \times 4 \sin^2 \theta \cos^2 \theta = 2 \times (4 \cos^2 \theta) = 1 - \cos 4\theta \Rightarrow \int_0^{\frac{\pi}{2}} 1 - \cos 4\theta = \left[\frac{\theta}{1} - \frac{\sin 4\theta}{4} \right]_0^{\frac{\pi}{2}} = \frac{\pi}{2}$

$$(2) \int_0^{\frac{\pi}{2}} \cos^2(2(\cos 4t)) dt = 2 \int_0^{\frac{\pi}{2}} \cos^2 u \cdot \sin u du \quad \cos u = u \quad -\sin u du = du$$

$$= 2 \int_0^1 u^2 du = 2 \cdot \left[\frac{u^3}{3} \right]_0^1 = \frac{2}{3}$$

$$(13): \int (\tan^2 \theta)^2 \tan \theta \sec^2 \theta d\theta = \int (\tan^4 \theta) \sec^2 \theta (\sin \theta \cos \theta) d\theta = \int (u^4 - u^2) u^2 du$$

$$= \int u^6 - u^4 du = \frac{1}{7} u^7 - \frac{1}{5} u^5 + C = \frac{1}{7} \tan^7 \theta - \frac{1}{5} \tan^5 \theta + C$$

$$(4): \int \frac{1}{\cos^2 x \sin^2 x} dx \quad \cos^2 x = 1 - \sin^2 x \quad \sin = u$$

$$(5) \quad \frac{1}{2} \left(\int_{-\pi/2}^{\pi/2} \cos(2x) \sin(x) dx + \int_{\pi/2}^{3\pi/2} \cos(2x) \sin(x) dx \right) = \frac{1}{2} \left(\int_{-\pi/2}^{\pi/2} \cos(2x) \sin(x) dx + \int_{\pi/2}^{3\pi/2} \cos(2x) \sin(x) dx \right) = \frac{1}{2} \left(\frac{1}{10} \sin(10x) + \left[-\frac{1}{5} \cos(5x) \right] \right) = \frac{1}{2} \left(\frac{1}{10} \sin(10x) - \frac{1}{5} \cos(5x) \right) = \frac{1}{20} \sin(10x) - \frac{1}{10} \cos(5x) + C$$

1. (1): $\int \sqrt{1-x^2} dx$ $x = \sin t$ $dx = \cos t dt$ $\Rightarrow x+C$
 $\int \sqrt{1-x^2} dx = \int \cos^2 t dt = \frac{1}{2} \int (1 + \cos 2t) dt = \frac{1}{2} t + \frac{1}{4} \sin 2t + C = \frac{1}{2} \arcsin x + \frac{1}{2} x \sqrt{1-x^2} + C$

(b) $z = \cos \theta$ da $\frac{dz}{d\theta} = -\sin \theta$ da $\frac{d\theta}{dz} = \frac{1}{-\sin \theta}$

$$\sqrt{3 - \frac{3}{2}}$$

(4) 2. Wert $\int_{-1}^1 x^2 dx = \frac{1}{3} x^3 \Big|_{-1}^1 = \frac{1}{3} (1 - (-1)^3) = \frac{1}{3} (1 + 1) = \frac{2}{3}$

$$1. \int \frac{1}{x^2+2x+1} dx = \int \frac{1}{(x+1)^2} dx = -\frac{1}{x+1} + C$$

$$\begin{aligned} \text{iv)} \quad & \frac{1}{2} \ln(x^2+2x) + \int \frac{-2}{x^2+2x} dx \Rightarrow -2 \left(\frac{1}{2} \ln|x| - \frac{1}{2} \ln|x+2| \right) \\ & = \frac{1}{2} \ln(x^2+2x) - \ln|x| + \ln|x+2| + C \end{aligned}$$

$$(3) \int -\frac{2}{x+5} dx + \int \frac{3}{x} dx + \int -\frac{1}{x^2} dx = -2 \ln|x+5| + 3 \ln|x| + \frac{1}{x} + C$$

$$(A)^{-1} = \frac{(A+B)(x^2+2) + (x+2)(x^2+1)}{(x^2+1)(x^2+2)} = \frac{(x^4+Bx^2+2Bx+2B) + (x^3+2x^2+x+2)}{(x^2+1)(x^2+2)}$$

$$\int \frac{3x-3}{x^2+1} dx = \int \frac{3x}{x^2+1} dx - \int \frac{3}{x^2+1} dx = \left[\frac{3}{2} \ln|x^2+1| - \frac{3}{x^2+1} \right] + C$$

(5)

$$1. \quad \lim_{x \rightarrow \infty} \int_0^x t^2 dt = \frac{1}{3} x^3 \quad \text{for } x \rightarrow \infty \quad \frac{d}{dx} = 2x \Rightarrow -\frac{1}{3} x^{-2} = -\frac{1}{3} \frac{1}{x^2} = -\frac{1}{3} (2x)^{-1} = -\frac{1}{6} (2x)^{-1} = -\frac{1}{6} \frac{1}{2x} = -\frac{1}{12x}$$

$$(2): \lim_{t \rightarrow \infty} \int_t^{\infty} \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{x} \right]_t^{\infty} = \lim_{t \rightarrow \infty} \left(0 + \frac{1}{t} \right) = 0$$

$$(3) \int_{-\infty}^{\infty} \frac{dx}{(2x^2+1)^2} : \frac{1}{4} \int_{-\infty}^{\infty} \frac{1}{(x^2+\frac{1}{2})^2} dx \quad x = \frac{1}{\sqrt{2}} \tan \theta \quad dx = \frac{1}{\sqrt{2}} \sec^2 \theta d\theta$$

$$= \frac{1}{4} \int_{-\pi/2}^{\pi/2} \frac{\sec^2 \theta}{\sec^2 \theta} d\theta = \frac{1}{4} [\theta]_{-\pi/2}^{\pi/2} = \frac{1}{4} \pi$$

$$(4) \quad \frac{1}{2} \int_{-1}^1 \frac{1}{x^2} - \frac{1}{x} dx = \frac{1}{2} \left(\int_{-1}^1 \frac{1}{x^2} dx - \int_{-1}^1 \frac{1}{x} dx \right) = \frac{1}{2} \left(\int_{-1}^1 \frac{1}{x^2} dx - \ln 3 \right)$$

2. $\sqrt{x} = u \quad du = \frac{dx}{2\sqrt{x}} \quad dx = 2u du$

$$\int e^{4x} \cdot 2 dx = -2e^{-4x} \cdot 2 \int e^{-4x} dx = -2e^{-4x} = -2e^{-4x} + C$$

$$\int_0^b e^{-x^2} dx = -\frac{1}{2} e^{-b^2} + \frac{1}{2} e^{-0^2} = -\frac{1}{2} e^{-b^2} + \frac{1}{2} e^0 = -\frac{1}{2} e^{-b^2} + \frac{1}{2}$$

[illegible]

7. $12x^2 y^3 z^3 \rightarrow \frac{1}{2}x^2 - y^4 - \frac{1}{10}z^2$ $S = \int_1^3 \sqrt{1 + \frac{1}{y^2}} dy = \int_1^3 \sqrt{1 + (y^2 - \frac{1}{y^2})^2} dy$
 $= \int_1^3 (y^2 + \frac{1}{y^2}) dy = [\frac{1}{3}y^3 - \frac{1}{y}]_1^3 = 9 - \frac{1}{3} - 1 + 1 = \frac{28}{3}$

$$2. \quad x = t, \quad y = \sqrt{1+t^2} + \arcsin t \quad x' = 1, \quad y' = \frac{t}{\sqrt{1+t^2}} + \frac{1}{\sqrt{1+t^2}} = \frac{1+t}{\sqrt{1+t^2}} \quad \int_0^{\pi} \frac{1+(1+t^2)}{1+t^2} dt \\ = \int_0^{\pi} \sqrt{1 + \frac{1+t^2}{1+t^2}} dt = \int_0^{\pi} \sqrt{1 + \frac{1}{1+t^2}} dt = \int_0^{\pi} \frac{1}{\sqrt{1+t^2}} dt + \int_0^{\pi} \sqrt{1+t^2} dt = 2\sqrt{2} \sqrt{1+t^2} - 4$$

3. 
$$= \frac{1}{2} \pi \cdot \frac{1}{2} + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1 - \frac{1 - e^{2i\theta}}{2}} \cdot i d\theta$$

$$= \frac{\pi}{4} + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} (1 + e^{2i\theta}) \cdot i d\theta = \frac{\pi}{4} + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos(2\theta)) \cdot i d\theta = \frac{\pi}{4} + \frac{i}{2} [\theta + \frac{1}{2} \sin(2\theta)]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= \frac{\pi}{4} + \frac{i}{2} [\ln(2) - \ln(1)] = \frac{\pi}{4} + \frac{i}{2} \ln(2)$$

$$b. \quad y = \frac{x^2}{16} + \frac{1}{2x^2}, \quad 1 \leq x \leq 2 \quad \int_a^b 2\pi y \, ds = \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} \, dx$$

$$\begin{aligned} f(x) &= \frac{1}{2} \pi \sqrt{\frac{2}{\pi}} x^{-\frac{1}{2}} \quad (f(0) = \frac{1}{2} \pi \sqrt{\frac{2}{\pi}} x^{-\frac{1}{2}} \Big|_{x=0} = \frac{1}{2} \pi \sqrt{\frac{2}{\pi}} \cdot 0 = 0) \\ |f(x)|^2 &= \pi \sqrt{\frac{2}{\pi}} x^{-\frac{1}{2}} \sqrt{\frac{2}{\pi}} x^{-\frac{1}{2}} = \pi \cdot \frac{2}{\pi} x^{-1} = 2x^{-1} = 2 \left(\frac{1}{2} x^2 \cdot \pi^{-1} \right)^{-\frac{1}{2}} \\ |f(x)|^2 &= \frac{1}{2} \pi \left(\frac{1}{x^2} \right)^{-\frac{1}{2}} = \frac{1}{2} \pi \left(\frac{x^{1+1}}{2} \right)^{-\frac{1}{2}} = \frac{1}{2} \pi \frac{2^{\frac{1}{2}}}{x^{\frac{1}{2}}} = \frac{1}{2} \pi \sqrt{2} \end{aligned}$$

$$2. \quad \oint_C \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{e^{i\theta}} i e^{i\theta} d\theta = i \int_0^{2\pi} 1 d\theta = 2\pi i$$

$$\begin{aligned} 3. \int_0^b 2\pi f(y) \sqrt{1+f'(y)^2} dy &= \int_0^b 2\pi \left(\frac{y}{b} \right) \sqrt{1+\left(\frac{y}{b}\right)^2} dy = \frac{2\pi}{b} \int_0^b y \sqrt{1+\frac{y^2}{b^2}} dy \\ &= \frac{2\pi}{b} \int_0^b \left(\frac{y^2}{b} + \frac{1}{2} \sqrt{b^2 + y^2} \right) dy = \frac{1}{b^2} \int_0^b (y^2 + \frac{1}{2} \sqrt{b^2 + y^2}) dy \\ &= \frac{1}{b^2} \left(\frac{y^3}{3} + \frac{1}{2} \sqrt{b^2 + y^2} \left(\frac{y}{2} + \frac{b^2}{2\sqrt{b^2 + y^2}} \right) \right) \Big|_0^b \end{aligned}$$

$$2. \text{ (iv) } \frac{1}{5+2^x} = \frac{1}{2^{x+1}+2^0} = \frac{2^0}{2^0+2^{x+1}} = \frac{1}{1+2^{x+1}} = \frac{1}{2^{x+1}+1}$$

(2) $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ $\frac{1}{1-\frac{1}{4}} = \frac{\frac{\pi^2}{6}}{1-\frac{1}{4}} = \frac{1}{\frac{3}{4}} = \frac{4}{3}$

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2022-493 4354

